# An Intelligent Assistant for Fertility Prediction,Wellness, and Support

Project ID – 25-26J-246

Project Proposal Report

Bachelor of Science (Hons) Degree in Information Technology
Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka (SLIIT)

August 2025

# Fertility Diet Recommendation

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August-2025

## DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

..... .....

Signature of the Supervisor Date

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..... 31/08/2025.

Signature of the Co-Supervisor Date

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**Abstract**

In Vitro Fertilization (IVF) Success Is Influenced by Multiple Biological and Lifestyle Factors, Among Which Nutrition Plays a Critical Yet Often Underemphasized Role. This proposal introduces an AI-driven personalized nutrition recommendation system designed to optimize IVF outcomes by aligning dietary intake with individual reproductive health needs. By analyzing key health parameters—such as hormone levels, BMI, and allergies—and integrating real-time data from wearable devices, the system generates dynamic, nutrient-rich diet plans tailored to support egg quality, hormonal balance, and uterine health. Key nutrients like folic acid, zinc, and omega-3 fatty acids are prioritized due to their established roles in embryo development, egg maturation, and implantation readiness. The system adapts dietary recommendations continuously based on daily physiological inputs, ensuring optimal nutritional support throughout the IVF process. This innovative approach positions personalized nutrition as a vital component of fertility care, enhancing the likelihood of successful implantation and pregnancy through targeted, data-informed dietary intervention.

**Keywords:** IVF, Personalized Nutrition, AI, Wearable Devices, Fertility

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## LIST OF ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| IVF | In Vitro Fertilization |
| AI | Artificial Intelligence |
| BMI | Body Mass Index |
| RDA | Recommended Daily Allowance |
| ML | Machine Learning |

| Abbreviation | Full Form |
|---|---|
| API | Application Programming Interface |
| Y/N | Yes/No |
| LKR | Sri Lankan Rupee |
| HRV | Heart Rate Variability |
| DB | Database |

# 1. INTRODUCTION

Infertility is a growing concern affecting millions of individuals and couples worldwide, and In Vitro Fertilization (IVF) has emerged as a leading solution in assisted reproductive technologies. While IVF offers a pathway to pregnancy, its success is influenced by a combination of clinical and lifestyle factors, including age, hormone levels, BMI, and overall health. Among these, one key element that is often underestimated is **nutrition**. Mounting evidence supports that what an individual consumes before and during IVF treatment can have a significant impact on the outcome.

Nutrients play a foundational role in reproductive health. For example, **folic acid** is critical for proper cell division and embryo development, **zinc** aids in hormonal regulation and egg maturation, and **omega-3 fatty acids** contribute to improved egg quality and a receptive uterine lining. These nutrients, among others, are essential for creating the optimal conditions for implantation and successful pregnancy. However, many standard IVF protocols overlook this dimension, and patients are often left with generic dietary advice that may not meet their specific biological needs.

Recognizing this gap, we propose a **personalized, AI-driven nutrition recommendation system** designed to complement IVF treatment by providing individualized dietary guidance. This system analyzes personal health data—including hormone levels, BMI, allergies, and other biomarkers—to generate tailored meal plans. These plans emphasize the intake of nutrient-dense foods like leafy greens, nuts, seeds, and fatty fish, which are aligned with the unique reproductive needs of each user.

What sets this system apart is its ability to adapt in **real-time**. By integrating with wearable health devices, it captures daily activity, sleep patterns, and physiological changes to continuously fine-tune nutritional recommendations. This dynamic feedback

loop ensures that the individual receives the right nutrients at the right time, making the diet not only personalized but also responsive. This leads to better hormonal balance, improved egg and embryo quality, and increased chances of successful implantation.

In conclusion, integrating personalized nutrition into the IVF process introduces a powerful, data-driven tool to support fertility outcomes. By combining AI technology with medical and nutritional science, this system fills a critical gap in current reproductive healthcare. It offers a proactive, patient-centered approach that maximizes the potential of IVF treatment, helping individuals achieve their dream of parenthood more effectively and naturally.

This report focuses specifically on **Component 2**, the AI-Driven Personalized Nutrition Recommendation Module, for which I, Narmatha Lingamayam, am responsible for development. The need for this component arises from two key challenges in the IVF process. First, it addresses the lack of individualized nutritional guidance in current fertility treatment protocols, which often overlook the critical role of diet in reproductive health. Without targeted nutritional support, patients may unknowingly follow generic diets that fail to meet their specific hormonal, metabolic, and reproductive needs, potentially affecting IVF success rates. Second, the module responds to the dynamic nature of each individual's physiology. Factors such as hormone fluctuations, BMI changes, and daily activity levels require dietary plans that can adapt in real time—something that static nutritional advice cannot offer.

Unlike traditional diet planning tools, this system leverages AI to generate **personalized, data-informed recommendations** based on each user's unique health profile. It integrates medical data (such as hormone levels and nutritional deficiencies) with real-time inputs from wearable devices, ensuring that nutrient intake is optimized at every stage of the IVF journey. This level of personalization and adaptability is uncommon in current fertility-support tools, which tend to treat nutrition as a secondary or fixed component.

In summary, this introduction highlights that the inclusion of a personalized AI nutrition module within an IVF support system represents a **forward-thinking innovation**. It bridges a significant gap in reproductive healthcare by combining nutrition science with real-time, AI-driven personalization. The sections that follow will explore the current role of nutrition in IVF success, review related technologies and existing limitations, outline the technical problem space, define the objectives of this component, and detail the methodology used in its development, followed by system requirements and references.

## 1.1 BACKGROUND AND LITERATURE SURVEY

### 1.1.1 Nutrition and IVF Success

In Vitro Fertilization (IVF) outcomes depend on multiple factors including maternal age, hormonal profile, lifestyle, and overall health. Recent studies emphasize the importance of nutrition as a critical, but often underappreciated, determinant of fertility outcomes. Essential nutrients such as **folic acid, zinc, vitamin D, antioxidants, and omega-3 fatty acids** have been shown to improve egg quality, support endometrial receptivity, and regulate hormonal balance. For example, **folic acid** promotes cell division and embryo development, **zinc** enhances oocyte quality and luteal function, while **omega-3 fatty acids** contribute to better implantation and embryo development (Cozzolino et al., 2023).

Literature also highlights that poor dietary habits, obesity, and micronutrient deficiencies negatively affect ovarian reserve, embryo quality, and implantation rates (Qureshi et al., 2024). Personalized nutrition, rather than a “one-size-fits-all” approach, is becoming increasingly important because women undergoing IVF differ in body composition, metabolic response, and lifestyle.

Advances in digital health now enable real-time tracking of nutrition and lifestyle data through wearable devices. By integrating this data with predictive analytics, personalized diet recommendations can be dynamically adjusted, improving adherence and maximizing IVF success.

### 1.1.2 Applications in Reproductive Medicine

Although personalized nutrition has been widely explored in general healthcare and chronic disease management, its application in reproductive medicine—particularly IVF—remains limited. Existing fertility apps mainly track menstrual cycles, ovulation windows, or provide generic lifestyle advice. Few systems incorporate **biometric, hormonal, and nutritional data** to create tailored interventions.

Analogous research in precision medicine and metabolic health demonstrates the effectiveness of **adaptive nutrition systems** that adjust recommendations in real time based on physiological changes. For IVF patients, this is especially relevant, as nutritional needs may vary across treatment phases (ovarian stimulation, egg retrieval, embryo transfer, and implantation support).

Thus, the integration of **AI-driven diet personalization** within IVF care represents a novel approach, bridging clinical nutrition, reproductive medicine, and digital health technologies.

### 1.1.3 Existing Techniques and Tools

Several approaches are currently studied for nutrition optimization in fertility:

- **Dietary Pattern Analysis**: Research shows that adherence to Mediterranean-style diets, rich in whole grains, fruits, fish, and healthy fats, improves IVF success rates.
- **Nutrient-Specific Studies**: Controlled trials link folic acid, vitamin D, and omega-3 supplementation to improved oocyte quality and higher implantation success.
- **Wearable Integration**: Devices tracking sleep, activity, and heart rate variability are increasingly used to correlate lifestyle with reproductive outcomes.

However, existing tools like **generic diet apps (e.g., MyFitnessPal, Cronometer)** or fertility apps (e.g., Flo, Clue) lack a targeted approach for IVF. Academic studies often stop at establishing correlations, without deploying **actionable real-time dietary interventions**.

## 1.2 RESEARCH GAP

Despite growing evidence linking nutrition with IVF outcomes, there is no widely implemented system that:

- Personalizes nutrition plans specifically for IVF patients.
- Dynamically adapts recommendations based on **real-time biometric and lifestyle data**.
- Integrates **clinical factors (hormones, BMI, allergies)** with **nutritional science**.
- Provides both **preventive and supportive guidance** during different IVF stages.

Most existing apps focus on cycle tracking or provide static nutrition tips. Academic literature identifies correlations but stops short of deploying applied solutions. Thus, a gap exists in creating a **comprehensive, adaptive, patient-centered nutrition system** for IVF support.

## 1.3 RESEARCH PROBLEM

The central research problem can be formulated as:
**How can we design a personalized, adaptive diet recommendation system that integrates nutritional science, patient health data, and real-time lifestyle inputs to improve IVF outcomes?**

This breaks into sub-questions:

1. What nutritional factors (macros, micros, diet patterns) most influence egg quality, hormonal balance, and implantation success?
2. How can we personalize diet plans based on patient-specific data (hormones, BMI, allergies, medical history)?
3. How can wearable device data (sleep, activity, stress levels) be integrated to dynamically adapt recommendations?
4. What is the optimal way to evaluate and measure the impact of personalized nutrition on IVF success rates?

# 2. OBJECTIVES

## 2.1 Main Objective

To develop a **personalized diet recommendation system** that enhances IVF success rates by analyzing patient health data, nutritional requirements, and lifestyle factors to deliver real-time adaptive nutrition guidance.

## 2.2 Specific Objectives

- **Nutrient Optimization**: Identify and prioritize nutrients (e.g., folic acid, zinc, omega-3s, vitamin D) most critical for reproductive health and IVF outcomes.
- **Personalization Engine**: Develop algorithms that generate individualized diet plans based on BMI, hormone levels, allergies, and clinical history.
- **Real-Time Adaptation**: Integrate wearable device data (activity, sleep, stress indicators) to dynamically adjust dietary recommendations.
- **User Experience & Compliance**: Provide meal suggestions, portion guidance, and reminders in an easy-to-use mobile interface, ensuring adherence.
- **Performance Evaluation**: Validate system effectiveness through simulation and/or pilot testing with real IVF patients, focusing on usability and improvement in predicted IVF success indicators.

# 3. Methodology

## 3.1 Data Collection

The system collects multiple types of data to build a personalized nutritional profile for women undergoing IVF treatment:

- **Clinical Data**: Hormone levels, BMI, medical history, and previous IVF outcomes.
- **Nutritional Data**: Recommended daily allowances, food composition databases, and nutrient–fertility correlations (e.g., folic acid, zinc, omega-3 fatty acids).
- **Wearable Data**: Daily activity levels, sleep quality, and stress indicators obtained from fitness trackers.
- **User Data**: Food preferences, allergies, and dietary restrictions provided through user input.

## 3.2 Data Processing and Analysis

The collected data is processed through a **hybrid recommendation model**:

- **Rule-Based Engine**: Encodes established nutritional guidelines from reproductive health studies (e.g., minimum folic acid intake for embryo development, zinc for egg health, omega-3 for implantation).
- **Machine Learning Component**: Learns user-specific responses to diet over time, identifying patterns such as improved sleep quality, better hormonal balance, or stable BMI.

This combination ensures both scientific accuracy and personalization.

## 3.3 Personalized Diet Recommendation

The system generates individualized meal plans that recommend foods rich in critical nutrients (e.g., leafy greens for folic acid, fish for omega-3s, nuts and seeds for zinc). Plans are further refined based on:

- **Real-Time Adaptation**: Wearable data integration allows dynamic adjustment of diet based on activity levels, stress, and sleep.
- **User Preferences**: Allergies, cultural dietary practices, and personal tastes are incorporated to ensure adherence.
- **Nutritional Balance**: Ensures that micronutrient and macronutrient requirements are met to optimize IVF success factors such as egg quality and uterine health.

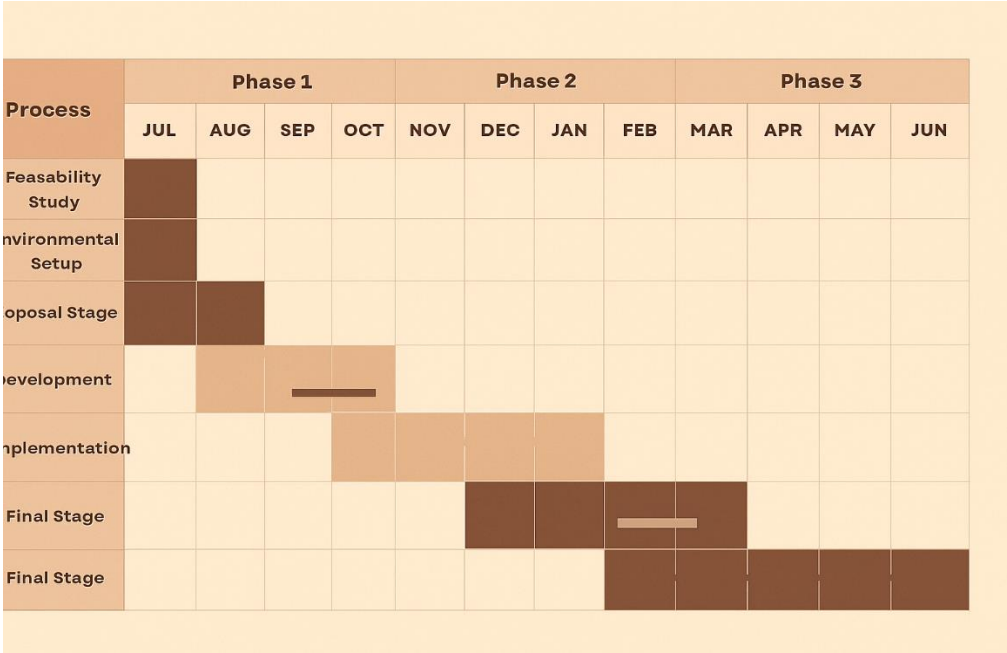
### 3.4 Validation and Evaluation

The effectiveness of the system is evaluated through:

- **Literature Validation**: Cross-checking nutrient recommendations with existing scientific studies on IVF outcomes.
- **Expert Review**: Consultation with nutritionists and fertility specialists to refine the diet rules.
- **Pilot Testing**: Implementing the system with a small group of users to monitor dietary adherence and track fertility-related health improvements.

## 4 GANTT CHART

*Table1: Gantt Chart*




# 5. Project Requirements

# 5. Functional Requirements

1. **Nutrition Data Collection**
    - The system must allow importing nutrition datasets from external sources (e.g., food composition databases, dietary references).

    - The system must store and organize data about meals, ingredients, nutrient values, allergies, and fertility-boosting properties.
    - The system must allow updating and editing nutrition data manually if new foods or ingredients are added.
2. **Wearable Device Integration**
    - The system must connect to wearable devices (smartwatches, fitness bands) via APIs.
    - The system must read relevant metrics from wearables (sleep, activity, heart rate) for personalized nutrition recommendations.
    - The system must sync nutrition intake data with wearable metrics to track adherence and provide real-time feedback.
3. **Data Processing and Analysis**
    - The system must process user input and wearable data to match them with nutrition datasets.
    - The system must generate personalized nutrition suggestions based on user health metrics and fertility goals.
4. **User Interaction**
    - The system must allow users to view daily/weekly meal plans and nutrient intake.
    - The system must provide alerts or suggestions if certain nutrient goals are not met.

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## Non-Functional Requirements

1. **Performance**
    - Data processing and synchronization with wearable devices should occur in real-time or near real-time (<5 seconds delay).
2. **Usability**
    - The interface for viewing nutrition plans and wearable integration should be intuitive, with clear labeling and easy navigation.
3. **Scalability**
    - The system must support adding new nutrition datasets or integrating additional wearable devices without major redesign.
4. **Reliability**
    - The system should ensure accurate storage and retrieval of nutrition and wearable data.
    - Failures in wearable data syncing must be logged and retried automatically.
5. **Security**
    - Sensitive user data (health metrics, dietary preferences) must be encrypted in storage and during transfer.
6. **Compatibility**

- The system must support multiple wearable devices and common operating systems (iOS, Android).

# 6 SYSTEM DIAGRAM

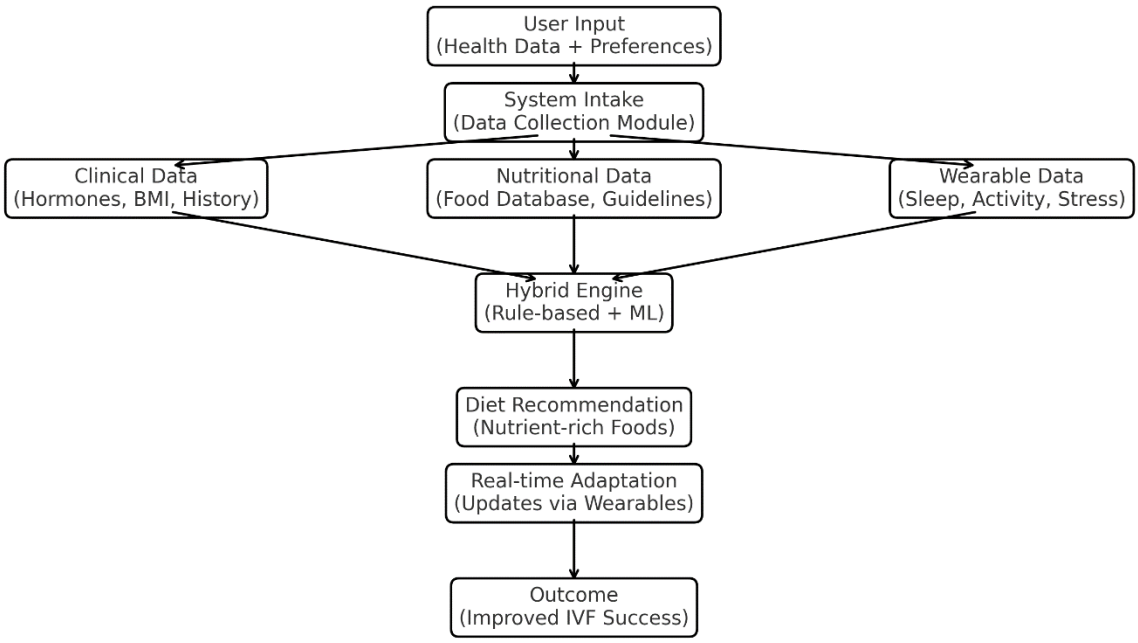


*Figure 1: System Diagram*

## System Diagram – IVF Nutrition & Diet Planning

**Components & Flow:**

1. **User Input Module**
   - Inputs: Age, BMI, allergies, dietary preferences, hormone levels, IVF history, lifestyle data.
   - Source: Manual entry + wearables (sleep, activity).
2. **Data Collection & Preprocessing**
   - Nutrition Dataset: Meals, ingredients, nutritional values, fertility-boosting nutrients.

- Hormone Response Dataset: How nutrients affect hormone levels.
- Preprocessing: Normalize, handle missing values, match user data with dataset.

3. **Recommendation Engine**
    - Rule-based logic: Minimum nutrients (folic acid, iron, omega-3).
    - ML personalization: Adjust meal plans based on user metrics & hormone responses.
    - Weekly Meal Planning: Suggest meals, alternatives, track intake.
4. **Monitoring Module**
    - Track daily nutrition adherence, wearable metrics (sleep, activity).
    - Feedback loop: Update recommendations dynamically.
5. **User Interface**
    - Displays: Daily/weekly meal plan, ingredient list, fertility-boosting tips, progress tracking.
    - Interactive: Suggest alternatives, notifications for missing nutrients.
6. **Evaluation & Reporting**
    - Generate summary reports: Nutrient intake, cycle-based adjustments, recommendations.

## 7. CONCLUSIONS

Nutrition plays a vital role in determining the success of In Vitro Fertilization (IVF). By ensuring that women receive the right balance of essential nutrients such as folic acid, zinc, and omega-3 fatty acids, overall reproductive health can be improved, egg quality can be enhanced, and the uterine environment can be better prepared for implantation. The proposed personalized diet recommendation system addresses this need by integrating clinical data, nutritional science, and real-time wearable inputs to deliver tailored meal plans. This approach not only supports hormonal balance and embryo development but also adapts dynamically to individual lifestyle factors, significantly increasing the chances of a successful IVF outcome.

## 8. Budget and Budget Justification

**Table 4: Budget and Budget Justification**

| Component | Amount (LKR) | Justification |
|---|---|---|
| Nutrition Data Collection | 9,000 | Covers costs for acquiring nutrition databases, food composition references, subscriptions, and any minor equipment or tools for accurate data collection. |
| Wearable Device Integration | 10,000 | Includes expenses for connecting nutrition data with wearable devices, software APIs, data syncing, and testing integration. |
| Miscellaneous | 4,000 | For communication, documentation, small equipment, and unforeseen costs. |
| **Total** | **23,000** | |

**Summary:**
The budget for the nutrition and wearable integration component of the IVF support system totals **LKR 23,000**. The allocation prioritizes **nutrition data collection** (LKR 9,000) to acquire comprehensive dietary information and relevant databases, and **wearable device integration** (LKR 10,000) to ensure seamless syncing of nutrition data with user health metrics. An additional **LKR 4,000** is reserved for miscellaneous expenses, covering communication, documentation, and unforeseen costs. This budget ensures the effective development and testing of the personalized diet recommendation system.

# 9. Commercialization

The development of the **IVF Nutritional Recommendation System**—integrating clinical, nutritional, wearable, and user-specific data—presents a significant commercial opportunity. This initiative addresses a **growing global need** for evidence-based, personalized health interventions that can improve fertility outcomes and overall reproductive health.

## 9.1 Market Opportunity

The potential market for this system extends well beyond IVF patients, covering multiple health and wellness domains:

- **Fertility Clinics and IVF Centers:** Globally, the demand for IVF and assisted reproductive technologies is rising. Clinics can adopt this system as an add-on

service to provide patients with personalized diet recommendations that improve treatment success rates.

- **Hospitals and Healthcare Providers:** Broader adoption is possible across general gynecology and nutrition departments to support maternal health, preconception care, and preventive health.
- **Wellness & Nutrition Apps:** The recommendation engine could be offered as a plugin or API for existing mobile health and nutrition platforms, allowing them to offer fertility-focused diet advice.
- **Corporate Healthcare Programs:** Employers, especially in sectors with wellness-driven CSR initiatives, could adopt the system to support reproductive health and employee well-being.
- **Global Digital Health Market:** Beyond IVF, the personalization engine can be extended to other contexts (e.g., weight management, diabetes, stress reduction), giving it long-term scalability.

With the rising prevalence of **delayed parenthood, lifestyle-related infertility, and increased awareness of nutritional interventions**, this system is well-positioned to meet a **critical gap in the digital health ecosystem**.

## 9.2 Revenue Model

A multi-channel revenue model ensures scalability and sustainability:

1. **Subscription Licensing for Clinics & Hospitals:** IVF centers and hospitals can subscribe to the system on a monthly or annual basis, with pricing tiers based on patient volume and feature usage.
2. **B2C Mobile App (Freemium Model):** A consumer-facing version could be launched, offering basic diet suggestions for free and premium features (detailed analytics, specialist consultations, advanced recommendations) under a subscription plan.
3. **API-as-a-Service:** The recommendation engine can be commercialized as an API, allowing third-party health and nutrition platforms to integrate fertility-focused dietary insights into their products.
4. **Partnerships with Nutrition & Supplement Companies:** Cross-promotions or integrated product recommendations can generate affiliate revenue streams while supporting patients' dietary needs.
5. **Grants & Research Collaborations:** Partnerships with universities, fertility research institutes, and global health organizations can provide early-stage funding and validation, while also expanding adoption.

This **diversified strategy** reduces reliance on a single customer base and creates opportunities for both **clinical adoption and direct consumer engagement**, positioning the IVF Nutritional Recommendation System as a scalable digital health innovation.

I. Female **Infertility and Diet: Is There a Role for a Personalized Nutritional Approach?**

- *Source*: Frontiers in Nutrition
- *Summary*: This narrative review discusses how personalized dietary interventions, including Mediterranean dietary patterns, can improve female reproductive health outcomes.
- *Link*: https://www.frontiersin.org/articles/10.3389/fnut.2022.927972/full

II. The **Role of Nutrient Supplements in Female Infertility: An Umbrella Review**

- *Source*: PubMed Central
- *Summary*: This umbrella review evaluates the effectiveness of various nutrient supplements in improving fertility outcomes for women with infertility.
- *Link*: https://pmc.ncbi.nlm.nih.gov/articles/PMC11722770/ Wiley Online Library+24Wiley Online Library+24Wiley Online Library+24

III. Female **Fertility and the Nutritional Approach: The Most Essential Nutrients for Reproductive Health**

- *Source*: PubMed Central
- *Summary*: This article highlights the essential nutrients that play a crucial role in female fertility, emphasizing the importance of a balanced diet in improving reproductive health and IVF outcomes.
- *Link*: https://pmc.ncbi.nlm.nih.gov/articles/PMC8634384/ Frontiers+11Prashanth Fertility Research Centre+11IVF Turkey+11

IV. Nutritional **Interventions for Enhancing Female Fertility: Evidence-Based Strategies**

- *Source*: Wiley Nutrition & Health Research
- *Summary*: This review discusses various nutritional interventions, including micronutrient supplementation, that can enhance female fertility by supporting oocyte quality and hormonal balance.
- *Link*: https://onlinelibrary.wiley.com/doi/10.1155/nrp/2137328

V. Nutritional **Supplements and IVF: An Evidence-Based Approach**

- *Source*: RBMO Journal / ScienceDirect

- *Summary*: This article systematically reviews the role of nutritional supplements in IVF treatments, providing evidence-based insights into how supplements can influence IVF success.
- *Link*: https://www.sciencedirect.com/science/article/abs/pii/S1472648323008696

VI. Exploring **Natural Supplements to Boost IVF Success**

- *Source*: IVF Turkey
- *Summary*: This article explores natural supplements, such as CoQ10 and omega-3 fatty acids, that have been shown to improve IVF success rates.
- *Link*: https://ivfturkey.com/exploring-natural-supplements-to-boost-ivf-success/ Birla Fertility & IVF+38IVF Turkey+38Frontiers+38

VII. Food **& You: A Digital Cohort on Personalized Nutrition**

- *Source*: PubMed Central
- *Summary*: This study implements a digital cohort to track food intake, physical activity, and gut microbiota, aiming to personalize nutrition for improved health outcomes.
- *Link*: https://pmc.ncbi.nlm.nih.gov/articles/PMC10688868/

VIII. Artificial **Intelligence in Personalized Nutrition and Food Manufacturing**

- *Source*: PubMed Central
- *Summary*: This article discusses the integration of AI in personalized nutrition, highlighting its potential in optimizing dietary interventions for reproductive health.
- *Link*: https://pmc.ncbi.nlm.nih.gov/articles/PMC12325300/ IVF Turkey+8Prashanth Fertility Research Centre+8Sunshine Egg Donor Agency+8

IX. Fertility **Tracking with Wearables: Empowering the Femtech Frontier**

- *Source*: Thryve Health Blog
- *Summary*: This blog post summarizes a JMIR review on how wearable technology can empower women to track fertility and make informed dietary and lifestyle choices.
- *Link*: https://thryvehealth.com/blog/fertility-tracking-with-wearables/

X. Meal **Meter: Using Multimodal Sensing and Machine Learning to Optimize Diet for Reproductive Health**

- *Source*: arXiv Preprint

- *Summary*: This preprint discusses the development of MealMeter, a system that uses multimodal sensing and machine learning to optimize diet for reproductive health.
- *Link*: https://arxiv.org/abs/2305.12233